The Midnight Express Mystery

Whodunnit? · Angles & Parallel Lines

Name: _____

The Midnight Express sped through the night with a famous passenger on board: the **Blue Diamond**, shining in its case. When the train went through a dark tunnel and came out the other side, the case was empty!

Six people were on board, each in a different car, and there are six ways to hide a diamond on a moving train. You are the train detective: find **who** took it, **where** they were, and **how** they hid it.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Conductor Hale Train Conductor	☐ Engine Car	☐ Hidden in a suitcase
☐ Porter Bell Luggage Porter	☐ Dining Car	☐ Sewn into a hat
☐ Madame Vesper Famous Singer	☐ Sleeping Car	☐ Dropped from the window
☐ Mr. Quint Jewel Dealer	☐ Baggage Car	☐ Locked in a violin case
☐ Dr. Marsh Traveling Doctor	☐ Observation Deck	☐ Folded in a newspaper
☐ Ms. Iris Dining Car Chef	☐ Caboose	☐ Buried in the coal bin

The culprit is _____ **in** _____ **using** _____

Show Your Work

The Midnight Express Mystery · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Midnight Express Mystery

Parallel lines are cut by a transversal. Alternate interior angles measure (3x + 10)° and (5x − 20)°. Find the measure of each angle.

A. 125°	The violin case was opened at the station. Just a violin and a spare string.
B. 15°	Dr. Marsh was checking the engineer’s pulse in the Engine Car. The engineer says he stayed the whole hour.
C. 110°	Conductor Hale was waving his lantern from the Caboose. The brakeman waved back.
D. 55°	Conductor Hale was driving the Engine Car. The engineer saw him the whole time.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Midnight Express Mystery

Parallel lines are cut by a transversal. Same-side interior angles measure (2x + 20)° and (3x − 10)°. Find the larger angle.

A. 34°	Porter Bell was sweeping out the Caboose. The pile of dust is still by the door.
B. 80°	The only violin case on board is dented and stuck shut. Nothing could be locked inside.
C. 88°	Dr. Marsh was sipping tea in the Dining Car. The waiter refilled his cup twice.
D. 92°	Every suitcase was searched at the station. None held the Diamond.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Midnight Express Mystery

Parallel lines are cut by a transversal. Corresponding angles measure (4x − 15)° and (2x + 25)°. Find the measure of each angle.

A. 65°	Porter Bell was sorting luggage in the Baggage Car with a helper beside him.
B. 115°	The violin case is lined with velvet and holds only sheet music.
C. 130°	Madame Vesper was practicing scales in the Caboose. The brakeman covered his ears.
D. 20°	Dr. Marsh was tucking a sleepy child in at the Sleeping Car. The mother thanked him.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Midnight Express Mystery

Parallel lines are cut by a transversal. Alternate exterior angles measure (6x − 4)° and (4x + 16)°. Find the measure of each angle.

A. 112°	Mr. Quint was polishing rings in the Caboose. A customer bought one right there.
B. 56°	The fancy hat was cut open and flattened. No diamond sewn inside.
C. 124°	The latch on the violin case is broken. It could not have been locked.
D. 10°	Dr. Marsh was lifting trunks in the Baggage Car. Porter Bell kept count of every one.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Midnight Express Mystery

The angles of a triangle measure 2x°, (3x + 5)° and (4x − 14)°. Find the largest angle.

A. 70°	Madame Vesper was resting in the Sleeping Car. Her maid never left her side.
B. 21°	Ms. Iris was hanging up her apron in the Caboose. The hook still holds it.
C. 42°	The violin case was weighed at the station. It was exactly as heavy as the violin alone.
D. 68°	Dr. Marsh was pointing out constellations on the Observation Deck. Three kids took notes.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Midnight Express Mystery

An exterior angle of a triangle measures (5x + 3)°. The two remote interior angles measure (2x + 30)° and (x + 21)°. Find the exterior angle.

A. 24°	Dr. Marsh was handing the engineer a cough drop in the Engine Car just before the whistle blew.
B. 123°	The windows were all latched shut during the trip, so nothing was dropped out.
C. 78°	The violin case rode in the overhead rack the whole trip. Its lock was never turned.
D. 57°	Conductor Hale was punching tickets in the Caboose. The stubs are stamped with the time.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Midnight Express Mystery

Two vertical angles measure (7x − 9)° and (5x + 17)°. Find the measure of each angle.

A. 164°	Porter Bell was napping on a bench in the Caboose. His snoring was heard down the track.
B. 98°	The violin case key was found in the lost-and-found box. No one used it that night.
C. 82°	Mr. Quint was admiring the stars from the Observation Deck with several passengers.
D. 13°	Dr. Marsh was carving roast beef in the Dining Car. Ms. Iris handed him the knife.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Midnight Express Mystery

Angles measuring (x + 40)° and (2x − 10)° are supplementary. Find the measure of the second angle.

A. 130°	The violin case smells only of rosin. Nothing else has ever been inside it.
B. 30°	Madame Vesper was gazing at the tracks from the Caboose. Her scarf blew off the back.
C. 50°	Dr. Marsh was reading by lamplight in the Sleeping Car. His bunkmate saw him until dawn.
D. 90°	The newspaper on the seat is a plain one with nothing folded inside.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Midnight Express Mystery

The vertex angle of an isosceles triangle measures 44°. Find the measure of each base angle.

A. 136°	Dr. Marsh was mending a torn suitcase in the Baggage Car. The needle and thread are still there.
B. 88°	The violin case was tapped and listened to. It sounded completely hollow.
C. 68°	Ms. Iris was serving dessert in the Dining Car. The passengers all remember it.
D. 46°	Mr. Quint was sorting his business cards in the Caboose. The brakeman took one.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

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Find the sum of the interior angles of a hexagon.

A. 720°	The coal bin was shoveled out. Only coal and soot.
B. 1080°	Dr. Marsh was watching the sunrise from the Observation Deck with a group of passengers.
C. 360°	The violin case was dusted for fingerprints. Only the owner’s were found.
D. 540°	Ms. Iris was stacking dessert plates in the Caboose. Not one plate was chipped.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.